See More, Know More: Unsupervised Video Object Segmentation with Co-Attention Siamese Networks

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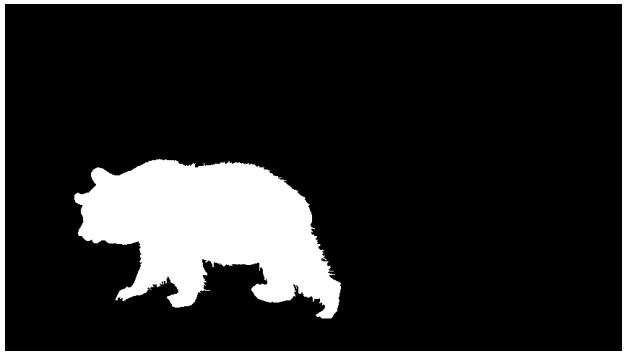
无监督视频语义分割

数据集：DAVIS2016 (前景和背景)



main contribution

- COSNet采用的训练方式是考虑一个pair，包含相同视频中的任意两帧，所以说极大的增加了数据量，不需要考虑时序关系，依次送入数据，而是可以打乱数据，随机组合。
- 显示建模帧和帧的联系，不依赖光流
- 统一的，端到端、可训练的高效网络

Train stage

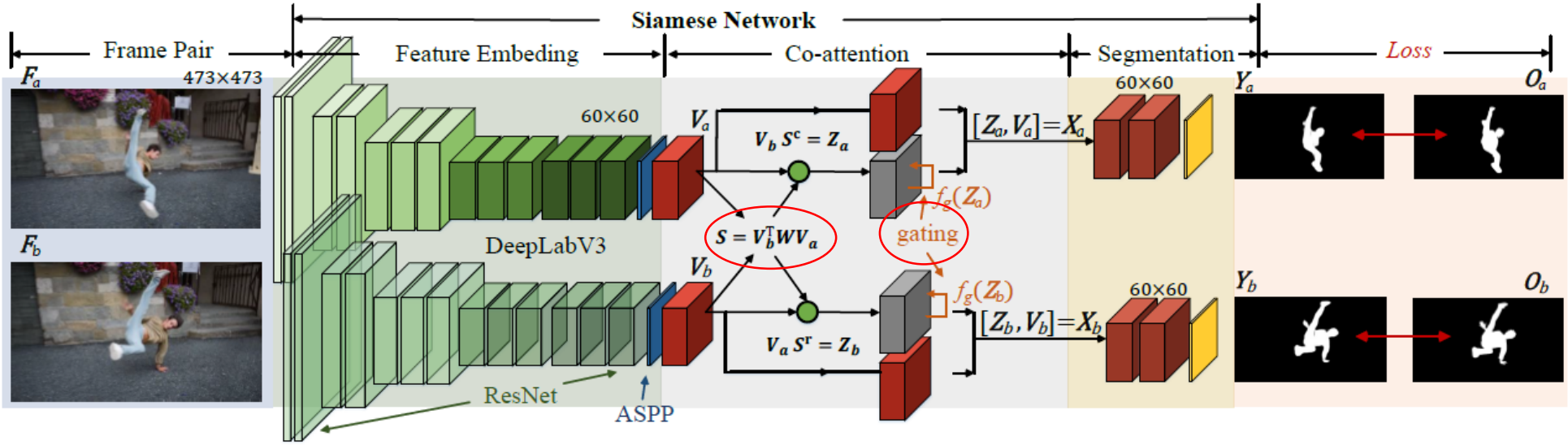


Figure 2. Overview of COSNet in the training phase. A pair of frames $\{\mathbf{F}_a, \mathbf{F}_b\}$ is fed into a feature embedding module to obtain the feature representations $\{\mathbf{V}_a, \mathbf{V}_b\}$. Then, the co-attention module computes the attention summaries that encode the correlations between $\mathbf{V}_a$ and $\mathbf{V}_b$. Finally, $\mathbf{Z}$ and $\mathbf{V}$ are concatenated and handed over to a segmentation module to produce segmentation predictions.

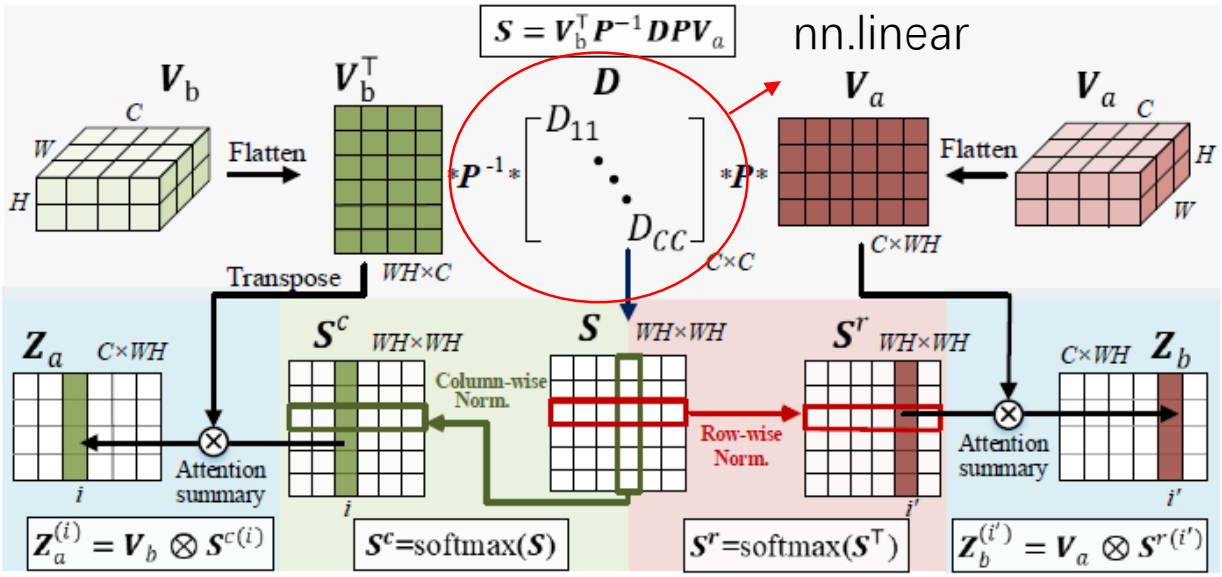


Figure 3. Illustration of our co-attention operation.

作者提出三种co-attention 变体（variants） 前文提到的 V_a, V_b默认被reshap为$R^{C \times HW}$

Vanilla co-attention

记仿射矩阵S，

$$S = V_b^T W V_a \in R^{WH \times WH}$$

$W \in R^{C \times C}$是需要学习的权重矩阵，S的每一个元素代表了两个特征中的两列的相似程度（向量乘积是一个标量）。把W对角化：

$$W = P^{-1} D P$$

P是可逆矩阵，S可以写成

$$S = V_b^T P^{-1} D P V_a$$

Symmetric co-attention (对称)

如果W是一个对称矩阵，那么W一定可以正定对角化，

$$S = V_b^T P^T D P V_a$$

$$S = (P V_b)^T D P V_a$$

P是正定矩阵，先将两个特征投影到正交空间中，然后在计算相似距离，这样子可以消除通道之间的联系，使得特征不是冗余的。

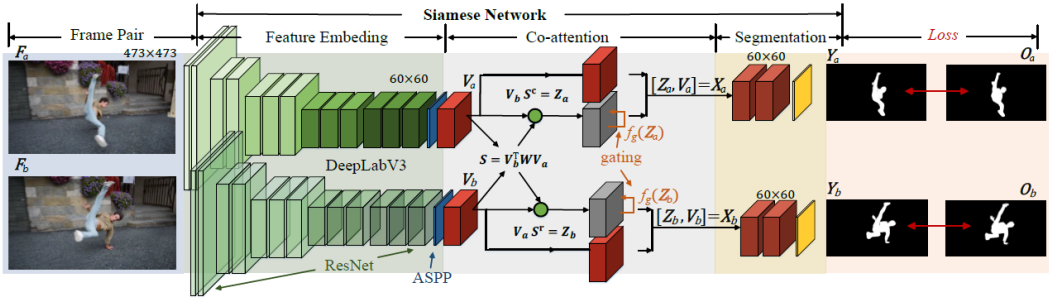
如果P是一个单位矩阵I，那么D可以被分解为两个对角矩阵。

$$S = V_b^T I^{-1} D I V_a = (D_a V_b)^T D_b V_a$$

那么一个对角矩阵和一个矩阵相乘，得到的结果其实是对角矩阵对角线的值乘以对应的列。可以视作增强了某一个通道

- **Gated co-attention**
- 作者引入Gated（门限），给学到的Z用同样大小的map reweights（空间注意力）

$$f_g(\mathbf{Z}_a) = \sigma(\mathbf{w}_f \mathbf{Z}_a + b_f) \in [0,1]^{WH},$$
$$f_g(\mathbf{Z}_b) = \sigma(\mathbf{w}_f \mathbf{Z}_b + b_f) \in [0,1]^{WH}, \quad (8)$$



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$$\mathbf{Z}_a = \mathbf{Z}_a \star f_g(\mathbf{Z}_a), \quad \mathbf{Z}_b = \mathbf{Z}_b \star f_g(\mathbf{Z}_b),$$

$$\mathbf{X}_a = [\mathbf{Z}_a, \mathbf{V}_a] \in \mathbb{R}^{W \times H \times 2C}, \quad \mathbf{X}_b = [\mathbf{Z}_b, \mathbf{V}_b] \in \mathbb{R}^{W \times H \times 2C}, \quad (10)$$

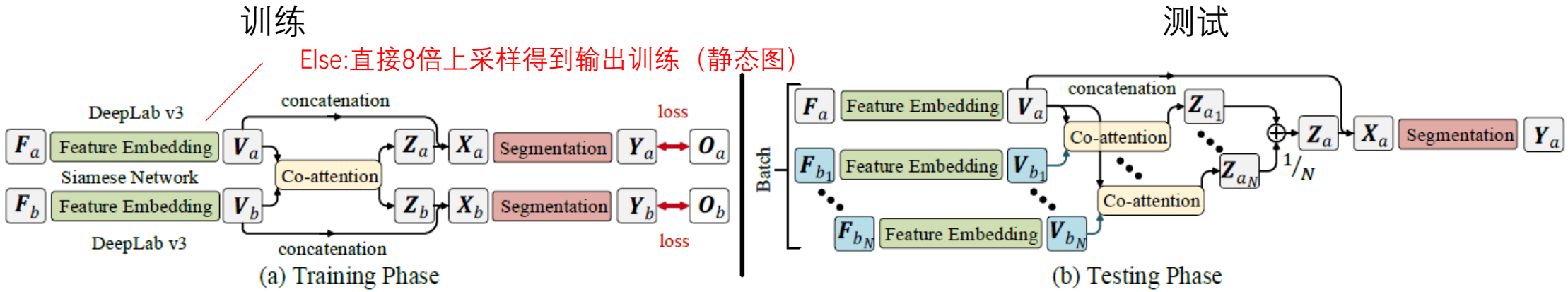
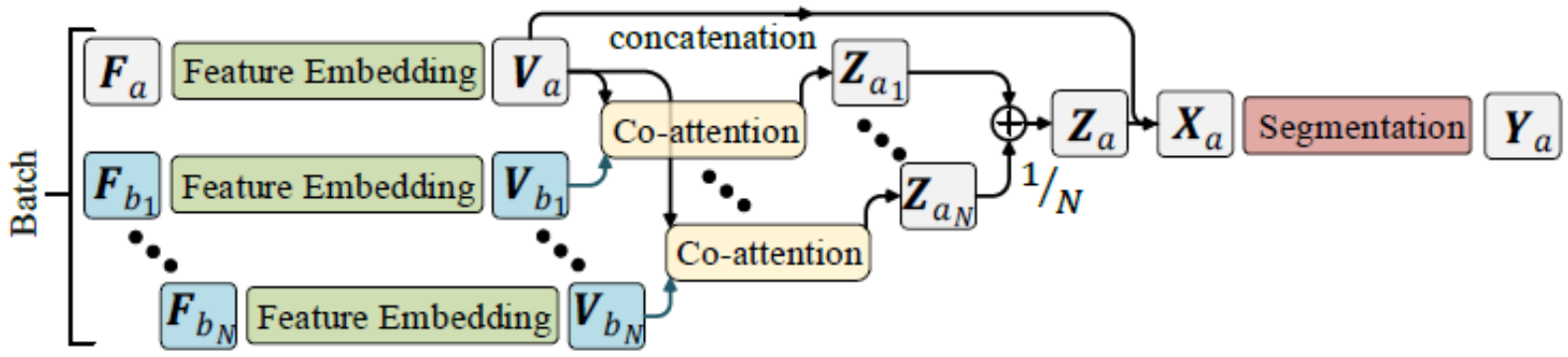


Figure 4. Schematic illustration of training pipeline (a) and testing pipeline (b) of COSNet.

训练（间隔训练）

If epoch%3!=0:随机从一个视频中选出两帧，经过backbone，经过co-attention,最后concat上backbone的特征，送到一个分割模块中，得到的预测结果用于计算损失函数。

else:作者提到使用在显著性数据集上预训练一个DeeplabV3，在deeplabV3后面加一个2通道的1x1的卷积作为输出。



(b) Testing Phase

首先选出N张参考帧集合$F_b = \{F_{b1} \ldots, F_{bn}\}$
和当前帧F_a，F_a和F_b中的所有参考帧都做成pair，得到一系列的Z_a，然后按照

$$\mathbf{Z}_a \leftarrow \frac{1}{N} \sum_{n=1}^{N} \mathbf{Z}_{a_n} \star f_g(\mathbf{Z}_{a_n}).$$

得到携带所有参考帧和当前帧联系的特征Z_a，然后concat Va,送到分割网络中。

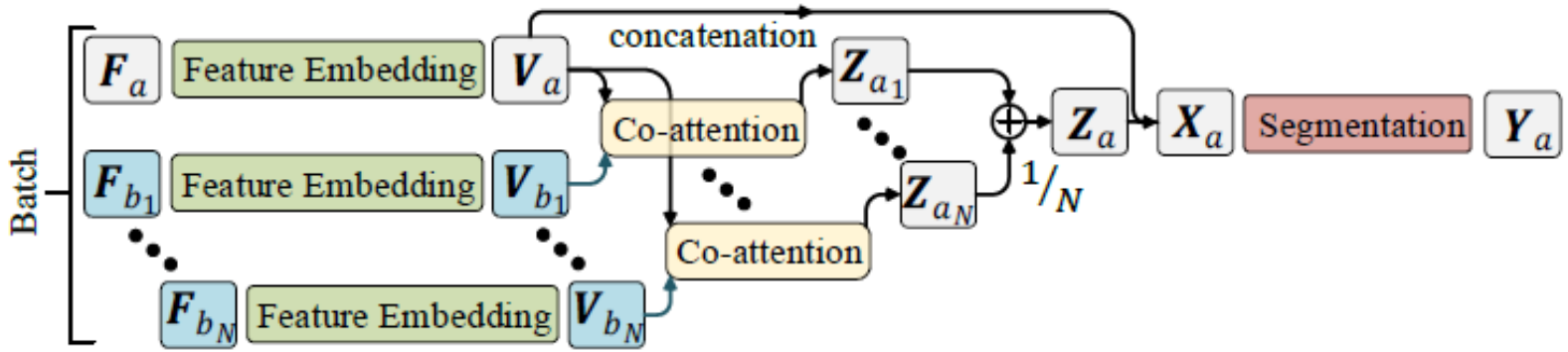
后处理使用了CRF

Network Variant	DAVIS		FBMS		Youtube-Objects	
	mean $\mathcal{J}$	$\Delta\mathcal{J}$	mean $\mathcal{J}$	$\Delta\mathcal{J}$	mean $\mathcal{J}$	$\Delta\mathcal{J}$
Co-attention Mechanism						
Vanilla co-attention (Eq. 3)	80.0	-0.5	75.2	-0.4	70.3	-0.2
Symmetric co-attention (Eq. 4)	80.5	-	75.6	-	70.5	-
Channel-wise co-attention (Eq. 5)	77.2	-3.3	72.7	-2.9	67.5	-3.0
w/o. Co-attention	71.3	-9.2	70.1	-5.5	62.9	-7.6
Fusion Strategy						
Attention summary fusion (Eq. 13)	80.5	-	75.6	-	70.5	-
Prediction segmentation fusion	79.5	-1.0	74.2	-1.4	69.9	-0.6
Frames Selection Strategy						
Global uniform sampling	80.53	-	75.61	-	70.54	-0.01
Global random sampling	80.52	-0.01	75.54	-0.02	70.55	-
Local consecutive sampling	80.26	-0.27	75.52	-0.09	70.43	-0.12

Table 1. Ablation study (§4.2) of COSNet on DAVIS16 [45], FBMS [41] and Youtube-Objects [47] datasets with different co-attention mechanisms, fusion strategies and sampling strategies.

$$\mathbf{Z}_a \leftarrow \frac{1}{N}\sum_{n=1}^{N} \mathbf{Z}_{a_n} \star f_g(\mathbf{Z}_{a_n}). \quad (13)$$

对于Xa取平均

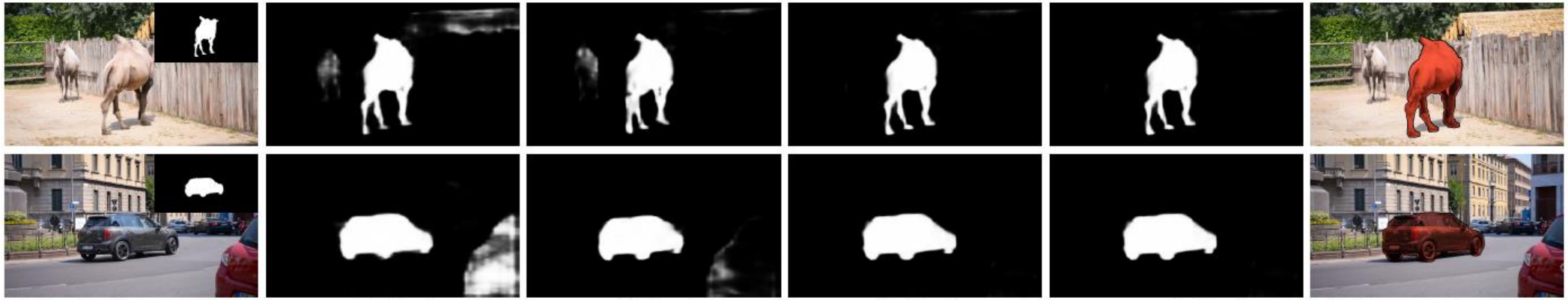


(b) Testing Phase

Dataset	Number of reference frames (N)				
	0	1	2	5	7
DAVIS	71.3	77.6	79.7	80.5	80.5
FBMS	70.2	74.8	75.3	75.6	75.6
Youtube-Objects	62.9	67.7	70.5	70.5	70.5

Table 2. Comparisons with different numbers of reference frames during the testing stage on DAVIS16 [45], FBMS [41] and Youtube-Objects [47] datasets (§4.2). The mean $\mathcal{J}$ is adopted.

result



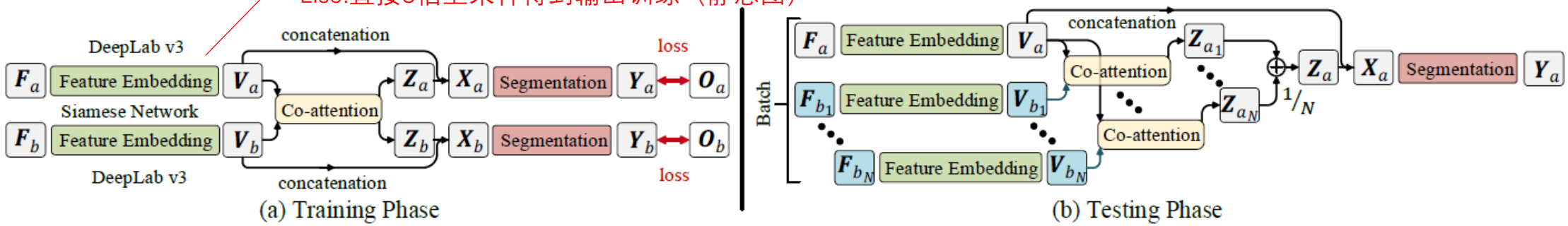
(a) Testing Frame with GT (b) $N = 0$ (c) $N = 1$ (d) $N = 2$ (e) $N = 5$ (f) Segmentation ($N = 5$)

Pytorch代码

训练

Else:直接8倍上采样得到输出训练（静态图）

测试



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